

Literature Citation Verification System Based on DeepScientist

Project A — Engineering Track: Evidence Chain Module

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STA304 Final Project · LLMs: From Principles to Applications

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The Problem

Core Pain Point

Research Agents (DeepScientist, OpenClaw) generate **professional-looking reports**, but users cannot answer three questions:

- ❶ **Which conclusions come from real sources?** — citations untraceable
- ❷ **Which are model inferences?** — no hallucination detection
- ❸ **Which have no evidence at all?** — report wording stronger than evidence

Our Goal

Build an **auditable, traceable evidence-chain system** that verifies every claim:

Upload PDF → Structured Evidence ID → API Verification → Scoring → Auto Audit

System Architecture

Pipeline Flow

Row 1 (Extraction)

QQ User $\longrightarrow$ crossdisc_idea (C) $\longrightarrow$ parse_pdf (A) $\longrightarrow$
verify_claim $\times N$ (A)

Row 2 (Scoring & Output)

score_batch (B) $\longrightarrow$ render_report (B) $\longrightarrow$ artifact_record +
memory_write (C) $\longrightarrow$ QQ Returns RYG Report

Three Roles

Person A — Tool Layer claim_extractor, semantic_verifier (PyPDF2,
S2/arXiv/Crossref API), HARD STOP RULE in builder.py

Person B — Scoring Layer traffic_light (7-branch RYG), render (Markdown),
MCP 5 tool registration, 6 Quest execution

FactCheck Pipeline Detail

Phase 0 — PDF Parsing

- PyPDF2 text extraction
- Regex-based reference section detection
- Split by [N] or [Author,Year] markers
- Unicode quote matching for title extraction
- Output truncation (200K chars / 40 claims max)

Phase 1 — Claim Verification $\times N$

- 3-tier fallback chain:
 1. Semantic Scholar (title+abstract)
 2. arXiv (Title API)
 3. Crossref (DOI/title search)
- Output: verdict + confidence + evidence snippet

Phase 2 — Batch Scoring (B)

Condition	Score	Signal
supported + conf ≥ 0.7	Green	●
supported + conf < 0.7	Yellow	●
not_found	Yellow	●
contradicted + conf ≥ 0.7	Red	●
contradicted + conf < 0.7	Yellow	●

Phase 3 — Render (B)

- Summary table (RYG counts)
- Per-claim detail cards
- Evidence Chain table (E00X $\rightarrow$ C00X)

Phase 4 — Persist (C)

- artifact__record $\rightarrow$ evidence_store
- memory__write $\rightarrow$ “episodes” kind

Evidence Chain & Enforcement Contract

3 Evidence Entry Types

Type	ID	Metadata
QQ Text	E001	text + SHA256
QQ Image	E001-img	dims + SHA256
QQ PDF	E001-pdf	pages + SHA256

Data Flow

Claim (C001) → VerificationResult → ScoredClaimResult → FactCheckResult

Dual-Layer Enforcement — Agent Cannot Bypass

Layer 1 — SKILL.md State Machine

Score → Render → Record → Memory → Deliver
Each state has exit checks; ≥ 1 fail = retry current state

Layer 2 — HARD STOP RULE (builder.py)

MUST call `score_batch` exactly once. No manual scoring.

MUST call `render_report`. No hand-written tables.

MUST call `artifact__record`, confirm status = ok.

MUST call `memory__write` after report generation.

HARD STOP: if any tool missing, continue execution

Test Results: 6 Disciplines × 6 Quests

Quest	Discipline	Paper Topic	Claims	verify	Score	Tools
027	ML/Federated	pFedAFM	40	85	17● 18● 5● FAIL	✓
028	Medical Edu.	Subinternship Dir.	8	27	1● 6● 1● FAIL	✓
029	Chemistry	DFT Clusters	1	7	1● 0● 0● PASS	✓
030	Optimization	SGD Convergence	40	105	10● 28● 2● FAIL	✓
031	Physics	LIGO PEMcheck	40	94	2● 37● 1● FAIL	✓
032	ML Survey	FL Survey	1	5	0● 1● 0● WARN	✓

Key Metrics

139 tests all passing | **996 events** | **323 verify_claim calls** | **36/36** mandatory tool calls completed

Failure Case Analysis

① Verifier False Positive (LIGO Q031)

- C036 (GW190707 event) flagged • contradicted
- **Cause:** Semantic Scholar matched a different paper with overlapping authors
- **Handling:** Report Caveats explicitly states “almost certainly a false negative”

② Abstract-Only Limitation (SGD Q030)

- C018/C031 mathematical claims unverifiable
- **Cause:** Theorems/lemmas require full-text, abstracts lack detail

③ PDF Parser Fallback (Medical Q028)

- `parse_pdf` returned 0 claims; Agent auto-fallback to `bash_exec`
- **Cause:** Medical papers indexed via PubMed, not arXiv — different citation format

④ Agent Bypassed Toolchain (mid-dev, fixed)

- Early runs: only 10/40 `verify_call`, **0 score_batch**, **0 render_report**
- **3-iteration fix:** MUST annotations → 5-State Machine → HARD STOP RULE
- **Result:** 36/36 compliance across all 6 quests

Boundary Cases & Three-Tier Annotation

6 Boundary Cases Covered

Case	Quest
Minimal claims (1)	Q029 (DFT PASS)
Max uncertainty (37/40)	Q031 (LIGO)
Cross-discipline refs	Q031 (astro-ph format)
Internal math proofs	Q027/030
Multi-citation [20] -- [24]	Q027
Fake ID detection	test_audit

Three-Tier Annotation

(Project core requirement)

- [E001] [E030] — Claims **supported** by evidence (● green)
- Inferred — **Logically inferred** but not directly proven by cited sources
- Needs Verification — **Insufficient evidence**; requires further experiment or manual review

`audit_report()` automatically scans for these annotations and cross-references with

Before / After Comparison

Dimension	Before (Original)	After (Our System)
Message Recording greenRYG!15 Citation Annotation	QQ logs only, no structure None	E00X evidence_store (16 fields) [E001] + RYG tri-color labels
Reference Verification greenRYG!15 Image Evi- dence	None No metadata	S2 / arXiv / Crossref API SHA256 + dimensions + format
PDF Evidence greenRYG!15 Audit Capa- bility	Not supported Impossible	Pages + SHA256 + full-text sidecar audit_report auto fake-ID detec- tion
Experiment Logging greenRYG!15 Test Cover- age	None 0	memory_write structured schema 139 tests / 100% pass
Grading Level	C+ (prompt only)	$\geq$ B- (independent MCP toolchain)

Current Limitations

- Abstract-only verification: most ● resolvable with full-text
- Verifier false positives: ~5–10% ● may be misclassifications
- PyPDF2 weak on scanned PDFs
- API-dependent: offline mode not supported
- No video/audio support yet

Future Improvements

- Full-text verification via arXiv open-access
- Local model fallback (Ollama + sentence-transformers)
- Structured NLP pipeline for claim extraction
- Multi-connector support (WeChat, Discord)
- Caching layer to avoid duplicate API calls

Team Contributions

Member	Role	Core Contribution
Miaoyi Liao	Person A (Tool)	claim_extractor, semantic_verifier, HARD STOP RULE, evidence-chain store
Xiaoyu Xiong	Person B (Scoring)	traffic_light (7 branches), render (Mark-down), MCP 5-tool registration, 6 Quest runs
Yuxiang Liu	Person C (Integration)	crossdisc_idea 5-Phase, QQ multi-modal evidence, daemon integration, memory schema

Cross-Collaboration Highlights

Phase 5 Enforcement: A (HARD STOP RULE) + B (5-State Machine) + C (memory/artifact schema) — three layers guarantee agent cannot bypass toolchain

Evidence Chain: A (storage + runner hooks) + B (audit + prompt protocol) + C (QQ multi-modal entry point) — full-link coverage

From “LLM Writes a Report” to “Agent Calls a Verifiable Toolchain”

Input	Verification	Output
6 disciplines	323 API lookups	RYG colored report
QQ real interaction	3-source fallback	Evidence Chain table
Text + Image + PDF	139 tests / 100%	Cross-disc. research ideas

Grade: $\geq$ **B-** (independent MCP toolchain, 36/36 mandatory calls)

https://github.com/leoplasture/STA304_Final_Project